Whole Genome Sequencing: Rapid Trio

Proband Report

BAYLOR GENETICS

DEMOGRAPHIC INFORMATION

PATIENT

NAME:	[redacted]
DATE OF BIRTH:	[redacted]
SEX:	[redacted]
MEDICAL RECORD #:	100567428
ACCESSION #:	
LAB NUMBER:	[redacted]
FAMILY NUMBER:	[redacted]

TEST INFORMATION

TEST NAME:	Rapid Trio WGS
TEST CODE:	1822
SAMPLE TYPE:	[redacted]
DATE COLLECTED:	[redacted]
DATE RECEIVED:	[redacted]
DATE REPORTED:	08/22/2025

RECIPIENT

PHYSICIAN NAME:	ELIZABETH AMES
FACILITY:	UNMI
LOCATION:	
PHONE:	- -
FAX:	- -

ADDITIONAL RECIPIENT

NAME:	Mallory Wagner
PHONE:	734-936-1129
FAX:	734-763-6561

ADDITIONAL RECIPIENT

NAME:	ADELYN BEIL
PHONE:	- -
FAX:	- -

CLINICAL INDICATION

Based on the submitted clinical information, the patient has microcephaly, highly arched eyebrow, long eyelashes, upslanted palpebral fissure, hearing impairment, anteverted nares, choanal atresia, cleft palate, microretrognathia, oral-pharyngeal dysphagia, chronic lung disease, hypoxemia, hypercapnia, respiratory failure, obstructive sleep apnea, cardiomyopathy, congestive heart failure, tetralogy of Fallot, pulmonary artery stenosis, ventricular septal defect, akinesia, edema, increased body weight, adrenal insufficiency, gastroesophageal reflux, split hand, 1–5 toe syndactyly, eczematoid dermatitis, and mottled pigmentation.

We have also received a sample from the father (DNA#3442531) and the mother (DNA#3442529) of this individual.

RESULTS

POSITIVE FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
Cornelia De Lange Syndrome 1	Autosomal Dominant	NIPBL: c.2479del, p.R827Gfs*20	Sequence Variant	Heterozygous	De Novo	Pathogenic
KMT2D-Related Disorders	Autosomal Dominant	KMT2D: c.1952C>G, p.S651W	Sequence Variant	Heterozygous	De Novo	Variant Of Uncertain Significance
BAZ2B-Related Neurodevelopmental Disorder	Autosomal Dominant	BAZ2B:HG38 chr2:159470492-159491775 DEL 21.28(kb)	Copy Number Variant	Heterozygous	Father	Variant Of Uncertain Significance
KCNQ1-Related Arrhythmias	Autosomal Dominant	KCNQ1: c.1552C>T, p.R518*	Sequence Variant	Heterozygous	N/A	Pathogenic

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RESULTS SUMMARY

A de novo heterozygous pathogenic variant in the NIPBL gene was detected, consistent with Cornelia de Lange syndrome 1.

A de novo heterozygous variant of uncertain significance in the KMT2D gene and a paternally inherited intragenic deletion in the BAZ2B gene were detected, possibly related to this patient's phenotype.

ADDITIONAL FINDINGS

ACMG recommended secondary findings (v3.2; PMID 37347242)

See table in Interpretation - ACMG Recommended Secondary Findings

Other Incidental Findings

None Detected.

Research Findings

None Detected.

RECOMMENDATIONS

• It is recommended that correlation of these findings with the clinical phenotype be performed.
• Genetic counseling is recommended.
• For sequence variants, testing is available for this individual's at risk relatives under test code 1580. Visit https://www.baylorgenetics.com/known-familial-mutations-policy/ for further information. Please consult with the lab for available test codes for other types of variants (CNV, STR, mito).
• Contact laboratory for requests regarding reanalysis and raw data.

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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
Cornelia De Lange Syndrome 1	Autosomal Dominant	NIPBL: c.2479del, p.R827Gfs*20	Sequence Variant	Heterozygous	De Novo	Pathogenic

GENE DESCRIPTION

The NIPBL gene encodes a protein forming a dimer with MAU2 as a cohesin loader complex that is essential for sister chromatid cohesion, chromosome condensation, and DNA repair.

DISEASE DESCRIPTION

Defects in the NIPBL gene may cause autosomal dominant Cornelia de Lange syndrome 1 (OMIM: 122470), which is a multisystem disorder characterized by growth retardation, developmental delay, intellectual disability, distinctive craniofacial features, hypertrichosis, seizures (less common), ocular anomalies, hearing loss, feeding difficulties, congenital heart defects (septal defects, pulmonary or peripheral pulmonary stenosis, coarctation of the aorta, tetralogy of Fallot), distal limb defects with greater involvement of the upper limbs (small hands/feet, phocomelia, oligodactyly, proximally placed thumbs, clinodactyly, toe syndactyly), gastrointestinal problems, genitourinary anomalies, and behavioral abnormalities such as autism and self-injurious behavior. Distinctive craniofacial features may include microbrachycephaly, low anterior hairline, synophrys, arched eyebrows, long eyelashes, ptosis, short nasal bridge with anteverted nares, cleft or high-arched palate, think upper lip, downturned corners of mouth, dental anomalies, and micrognathia.

VARIANT DETAILS - NIPBL: c.2479del, p.R827Gfs*20

- (GRCh38)chr5:36985658CA>C - NM_133433.4
- The c.2479del (p.R827Gfs*20) variant in the NIPBL gene has been described as pathogenic in ClinVar (ID: 950707). This variant has been previously reported as de novo in individuals with Cornelia de Lange syndrome (PMID: 20824775, 36913886).
- This variant has not been observed in the gnomAD v4 dataset.
- This frameshift variant is located in exon 10 of 47 and predicted to be deleterious.

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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
KMT2D-Related Disorders	Autosomal Dominant	KMT2D: c.1952C>G, p.S651W	Sequence Variant	Heterozygous	De Novo	Variant Of Uncertain Significance

GENE DESCRIPTION

The KMT2D gene encodes a histone methyltransferase that methylates the lysine 4 of histone H3 (H3K4) as part of the chromatin remodeling machinery involved in transcriptional regulation and DNA repair.

DISEASE DESCRIPTION

Defects in the KMT2D gene may cause autosomal dominant Kabuki syndrome 1 (OMIM: 147920), which is characterized by postnatal growth restriction, developmental delay, intellectual disability, distinct facial features (long palpebral fissures, eversion of the lateral third of the lower eyelids, wide-arched eyebrows, large prominent/cupped earlobes, short columella with broad/depressed nasal tip), infantile hypotonia, persistence of fingertip pads, and musculoskeletal anomalies (joint hypermobility, scoliosis, brachydactyly, short fifth finger). Additional clinical features may include seizures, ptosis, strabismus, ear pits, hearing loss, cleft lip/palate, widely spaced teeth, hypodontia, congenital heart defects, congenital hypothyroidism, gastrointestinal and genitourinary anomalies such as anal atresia and horseshoe kidneys, an increased susceptibility to infections, autoimmune disorders, and endocrinologic abnormalities (hyperinsulinism, premature thelarche). Defects in the KMT2D gene may also cause autosomal dominant branchial arch abnormalities, choanal atresia, athelia, hearing loss, and hypothyroidism syndrome (OMIM: 620186), which is characterized by choanal atresia, athelia or hypoplastic nipples, branchial sinus abnormalities, neck pits, lacrimal duct anomalies, hearing loss, external ear malformations, and thyroid abnormalities. Additional clinical features may include growth retardation, developmental delay, and intellectual disability.

VARIANT DETAILS - KMT2D: c.1952C>G, p.S651W

- (GRCh38)chr12:49051731G>C - NM_003482.4
- The c.1952C>G (p.S651W) variant in the KMT2D gene has not been described in ClinVar.
- This variant has not been observed in the gnomAD v4 dataset.
- This missense variant occurs at a residue with low evolutionary conservation, and its effect is predicted to be inconclusive (REVEL = 0.211, AlphaMissense = 0.239, CADD = 20.4).

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INTERPRETATION - PHENOTYPE RELATED FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
BAZ2B-Related Neurodevelopmental Disorder	Autosomal Dominant	BAZ2B:HG38 chr2:159470492-159491775 DEL 21.28(kb)	Copy Number Variant	Heterozygous	Father	Variant Of Uncertain Significance

GENE DESCRIPTION

The BAZ2B gene encodes a member of the bromodomain family and is potentially involved in transcriptional activation through chromatin remodeling.

DISEASE DESCRIPTION

Defects in the BAZ2B gene may cause autosomal dominant BAZ2B-related neurodevelopmental disorder (PMID: 37872713, 31999386), which is characterized by developmental delay, intellectual disability, and autism spectrum disorder. Additional clinical features are highly variable, including growth delay, nonspecific craniofacial dysmorphism often with macrocephaly or microcephaly, epicanthal folds, and small ears, abnormal brain imaging, vision issues (strabismus, myopia, hyperopia), gastrointestinal problems, congenital heart defects, and musculoskeletal anomalies. Currently, the gene-disease association between BAZ2B and neurodevelopmental defects is supported by a limited number of affected individuals. Further clinical and genetic studies are warranted to better define disease onset, penetrance, and genotype-phenotype correlations.

VARIANT DETAILS - BAZ2B:HG38 chr2:159470492-159491775 DEL 21.28(kb)

- (GRCh38)chr2:159470491-159491774DEL -
- This copy number loss is mapped to NM_013450.4:c.-2-13051_145+8086del, including the exon 3 (the first coding exon) of the BAZ2B gene.
- The exon 3 deletion in the BAZ2B gene has not been described in ClinVar.
- The exon 3 deletion has not been observed in gnomAD.
- A start-loss variant, c.2T>C (p.M1?) in the BAZ2B gene, has been reported as de novo in an individual with neurodevelopmental defects (PMID: 37872713).

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INTERPRETATION - ACMG RECOMMENDED SECONDARY FINDINGS

DISEASE	INHERITANCE PATTERN	GENE / VARIANT	VARIANT TYPE	GENOTYPE	INHERITED FROM	VARIANT CLASSIFICATION
KCNQ1-Related Arrhythmias	Autosomal Dominant	KCNQ1: c.1552C>T, p.R518*	Sequence Variant	Heterozygous	N/A	Pathogenic

GENE DESCRIPTION

The KCNQ1 gene encodes a voltage-gated potassium channel that functions in the repolarization phase of the cardiac action potential.

DISEASE DESCRIPTION

Defects in the KCNQ1 gene may cause autosomal dominant long QT syndrome 1 (OMIM: 192500), which is characterized by a prolonged QT interval and polymorphic ventricular arrhythmias (torsade de pointes), leading to recurrent syncope, seizures, or sudden death. Defects in the KCNQ1 gene may also cause autosomal dominant short QT syndrome 2 (OMIM: 609621), which is characterized by short QT intervals and episodes of bradycardia, atrial fibrillation, leading to recurrent syncope, seizures, or sudden death. Defects in the KCNQ1 gene may also cause autosomal dominant familial atrial fibrillation 3 (OMIM: 607554), which is characterized by episodes of rapid/irregular heart beats, atrial fibrillation, syncope, or sudden death without demonstrable cardiac or noncardiac causes to account for the episode. The disease mechanism underlying KCNQ1-related arrhythmias has been proposed to be loss-of-function or dominant-negative for long QT syndrome 1 and gain-of-function for short QT syndrome 2 and familial atrial fibrillation 3 (PMID: 9323054, 12522251, 35806392).

VARIANT DETAILS - KCNQ1: c.1552C>T, p.R518*

- (GRCh38)chr11:2768881C>T - NM_000218.3
- The c.1552C>T (p.R518*) variant in the KCNQ1 gene has been described as pathogenic in ClinVar (ID: 3131). This variant has been previously reported in individuals with Long QT syndrome 1 (PMID: 23098067, 24552659).
- This variant has been observed in gnomAD with a frequency of < 0.05%.
- This stop-gain variant is located in exon 12 of 16 and predicted to be deleterious.

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QUALITY METRICS

Percentage depth of coverage at 20x: **99.74%**
Mean depth of coverage: **61.58x**

METHODOLOGY

Genomic DNA was fragmented and indexed to prepare individual libraries. The libraries were purified and pooled for sequence analysis on the Illumina NovaSeq platform for paired end reads. Internal quality control measures are performed to ensure sample integrity.

Data analysis and interpretation are performed by the Baylor Genetics analytics pipeline. Data are aligned to the human reference genome build GRCh38 using the Illumina Dragen BioIT Platform. Variant calling is performed using the Illumina Dragen haplotype-based variant calling system and Illumina Dragen genome wide depth based CNV (Copy Number Variants) caller with custom modifications from Baylor Genetics. Short tandem repeat calling is performed using the Illumina Expansion Hunter tool.

The automated genetic interpretation platform from Emedgene is implemented for variant annotation and interpretation with proprietary algorithms and open-source data sets such as gnomAD, EVS (Exome Variant Server), ClinVar, and professional resources such as HGMD (Human Gene Mutation Database) Pro. The variants were interpreted according to ACMG (American College of Medical Genetics) guidelines (reference 1) and patient phenotypes.

Variants are confirmed by Sanger sequencing, specific PCR, Multiplex Ligation-dependent Probe Amplification (MLPA), or another methodology, if sequencing or copy number variants detected by NGS do not meet internal quality standards or are within highly homologous regions.

Synonymous variants, noncoding variants not affecting splicing sites, and common benign variants are excluded from interpretation unless previously reported in the literature as possibly pathogenic variants. This test may not provide detection of certain variants or portions of certain genes due to insufficient sequencing depth, local sequence characteristics, high/low genomic complexity, or the presence of closely related pseudogenes. Small deletions or insertions larger than 25bp, low-level mosaic variants, structural variants such as inversions, and/or balanced translocations may not be detected with this technology.

Variant types being analyzed include sequencing variants and CNV in nuclear genes and mitochondrial DNA (mtDNA), and short tandem repeats (STR). Uniparental disomy (UPD) analysis is done only in a trio setting. Regions of homozygosity (ROH) are reported when greater than 5 Mb. Sensitivity of detection is decreased for mtDNA sequencing variants and CNVs at heteroplasmy levels lower than 5% and 10%, respectively. Pathogenic events of short tandem repeats can be detected within the genomic regions of following genes: AFF2, AR, ATN1, ATXN1, ATXN2, ATXN3, ATXN7, ATXN10, ATNX8OS, C9orf72, CACNA1A, CNBP, CSTB, DIP2B, DMPK, FGF14, FMR1, FXN, GLS, HTT, JPH3, NOP56, NOTCH2NLC, PABPN1, PHOX2B, PPP2R2B, RFC1, TBP, and TCF4. Sensitivity of detection is reduced for borderline or incomplete penetrance alleles.

The proband report contains results of genes related to the patient's clinical phenotype. Variants relevant to the phenotype are reported as positive (pathogenic/likely pathogenic) or variants of uncertain significance (VUS). Benign and likely benign variants are not reported. This interpretation is based on the clinical information provided by the referring institution and our current understanding of the gene and variants at the time of reporting. Additional methodologies may be used to aid in variant interpretation. Results in the report should be carefully correlated to the individual's clinical findings by the referring physician.

If elected, secondary findings including pathogenic and likely pathogenic variants in genes included in the ACMG secondary findings recommendations, or other findings determined to be severe and medically significant (incidental findings), will be reported. The ACMG recommendations (ACMG SF v3.2 (reference 2)) include the following genes: ACTA2, ACTC1, ACVRL1, APC, APOB, ATP7B, BAG3, BMPR1A, BRCA1, BRCA2, BTD, CACNA1S, CALM1, CALM2, CALM3, CASQ2, COL3A1, DES, DSC2, DSG2, DSP, ENG, FBN1, FLNC, GAA, GLA, HFE, HNF1A, KCNH2, KCNQ1, LDLR, LMNA, MAX, MEN1, MLH1, MSH2, MSH6, MUTYH, MYBPC3, MYH11, MYH7, MYL2, MYL3, NF2, OTC, PALB2, PCSK9, PKP2, PMS2, PRKAG2, PTEN, RB1, RBM20, RET, RPE65, RYR1, RYR2, SCN5A, SDHAF2, SDHB, SDHC, SDHD, SMAD3, SMAD4, STK11, TGFBR1, TGFBR2, TMEM127, TMEM43, TNNC1, TNNI3, TNNT2, TP53, TPM1, TRDN, TSC1, TSC2, TTN, TTR, VHL, and WT1. Variants unrelated to the patient's phenotype but potentially clinically significant in genes with no known disease association will be reported if elected.

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LIMITATIONS

The interpretation of nucleotide changes is based on our current understanding of the specific genes. This interpretation may change over time as more information about these genes becomes available. Possible diagnostic errors include sample mix-ups, genetic variants that interfere with analysis, incorrect assignment of biological parentage, or other sources. Please contact a genetic counselor at Baylor Genetics if there is reason to suspect one of these sources of error.

REFERENCES

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015 May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.
2. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, Klein TE, Richards CS, Stewart DR, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2023 Aug;25(8):100866. doi: 10.1016/j.gim.2023.100866. PMID: 37347242.

DISCLAIMER

This test was developed, and its performance was determined by the Baylor Miraca Genetics Laboratory DBA Baylor Genetics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). Since FDA is not required for clinical use of this test, this laboratory has established and validated the test's accuracy and precision, pursuant to the requirement of CLIA'88. This laboratory is licensed and/or accredited under CLIA and CAP (CAP#2109314 / CLIA# 45D0660090; Lab Director: Christine M. Eng, MD).

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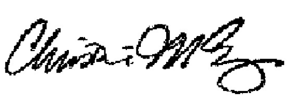

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